공학수학(2)

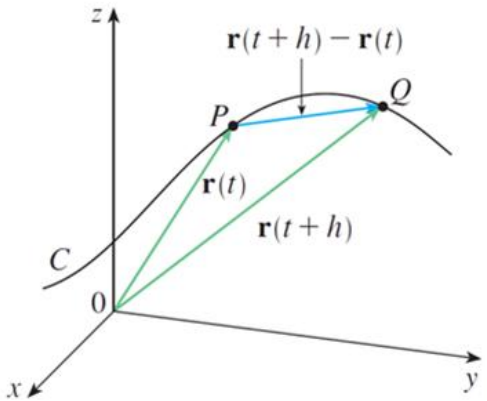
12.2 Derivatives and Integrals of Vector Functions

Derivatives

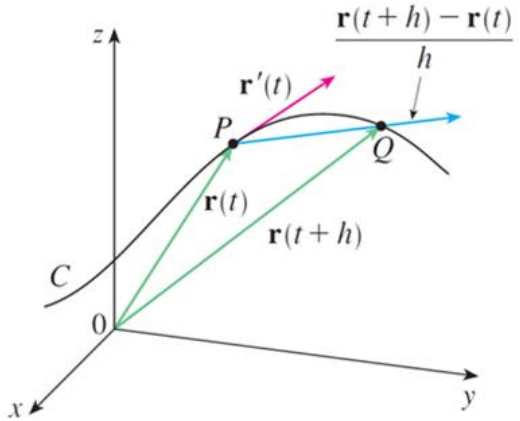
The derivative $\mathbf{r}'$ of a vector function $\mathbf{r}$ is defined in much the same way as for real-valued functions:

$$\mathbf{1} \quad \frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

if this limit exists. The geometric significance of this definition is shown in Figure 1.



(a) The secant vector $\overrightarrow{PQ}$



(b) The tangent vector $\mathbf{r}'(t)$

Figure 1

Derivatives

If the points P and Q have position vectors $\mathbf{r}(t)$ and $\mathbf{r}(t + h)$, then $\overrightarrow{PQ}$ represents the vector $\mathbf{r}(t + h) - \mathbf{r}(t)$, which can therefore be regarded as a secant vector.

If $h > 0$, the scalar multiple $\left(\frac{1}{h}\right)(\mathbf{r}(t+h)-\mathbf{r}(t))$ has the same direction as $\mathbf{r}(t + h) - \mathbf{r}(t)$. As $h \to 0$, it appears that this vector approaches a vector that lies on the tangent line.

For this reason, the vector $\mathbf{r}'(t)$ is called the **tangent vector** to the curve defined by $\mathbf{r}$ at the point P, provided that $\mathbf{r}'(t)$ exists and $\mathbf{r}'(t) \neq \mathbf{0}$.

The **tangent line** to C at P is defined to be the line through P parallel to the tangent vector $\mathbf{r}'(t)$.

Derivatives

The following theorem gives us a convenient method for computing the derivative of a vector function **r**: just differentiate each component of **r**.

2 Theorem If $\mathbf{r}(t) = \langle f(t),\ g(t),\ h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where *f*, *g*, and *h* are differentiable functions, then

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t)\,\mathbf{i} + g'(t)\,\mathbf{j} + h'(t)\,\mathbf{k}$$

A unit vector that has the same direction as the tangent vector is called the **unit tangent vector T** and is defined by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Example 1

(a) Find the derivative of $\mathbf{r}(t) = \left(1 + t^3\right)\mathbf{i} + te^{-t}\mathbf{j} + \sin 2t\ \mathbf{k}$.

(b) Find the unit tangent vector at the point where $t = 0$.

Solution:

(a)

(b)

Derivatives

Just as for real-valued functions, the **second derivative** of a vector function **r** is the derivative of $\mathbf{r}'$, that is, $\mathbf{r}'' = (\mathbf{r}')'$.

For instance, the second derivative of the function, $\mathbf{r}(t) = \langle 2\cos t, \sin t, \quad t \rangle$, is

$$\mathbf{r}''(t) = \langle -2\cos t, -\sin t, 0 \rangle$$

Differentiation Rules

The next theorem shows that the differentiation formulas for real-valued functions have their counterparts for vector-valued functions.

3 Theorem Suppose **u** and **v** are differentiable vector functions, c is a scalar, and f is a real-valued function. Then

1. $\dfrac{d}{dt}\big[\mathbf{u}(t)+\mathbf{v}(t)\big]=\mathbf{u}'(t)+\mathbf{v}'(t)$

2. $\dfrac{d}{dt}[c\mathbf{u}(t)]=c\mathbf{u}'(t)$

3. $\dfrac{d}{dt}\big[f(t)\mathbf{u}(t)\big]=f'(t)\mathbf{u}(t)+f(t)\mathbf{u}'(t)$

4. $\dfrac{d}{dt}\big[\mathbf{u}(t)\cdot\mathbf{v}(t)\big]=\mathbf{u}'(t)\cdot\mathbf{v}(t)+\mathbf{u}(t)\cdot\mathbf{v}'(t)$

5. $\dfrac{d}{dt}\big[\mathbf{u}(t)\times\mathbf{v}(t)\big]=\mathbf{u}'(t)\times\mathbf{v}(t)+\mathbf{u}(t)\times\mathbf{v}'(t)$

6. $\dfrac{d}{dt}\big[\mathbf{u}\big(f(t)\big)\big]=f'(t)\mathbf{u}'(f(t))$ (Chain Rule)

Differentiation Rules

We use Formula 4 to prove the following theorem.

4 Theorem if $|\mathbf{r}(t)| = c$ (a constant), then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t.

PROOF

Since

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = |\mathbf{r}(t)|^2 = c^2$$

and c^2 is a constant, Formula 4 of Theorem 3 gives

$$0 = \frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] = \mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 2\mathbf{r}'(t) \cdot \mathbf{r}(t)$$

Thus $\mathbf{r}'(t) \cdot \mathbf{r}(t) = 0$, which says that $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$.

Differentiation Rules

Geometrically, this result says that if a curve lies on a sphere with center the origin, then the tangent vector $\mathbf{r}'(t)$ is always perpendicular to the position vector $\mathbf{r}(t)$. (See Figure 4.)

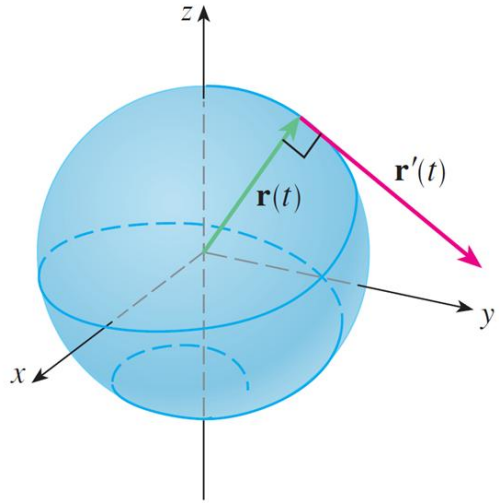


Figure 4

Integrals

The **definite integral** of a continuous vector function **r**(*t*) can be defined in much the same way as for real-valued functions except that the integral is a vector.

But then we can express the integral of **r** in terms of the integrals of its component functions *f*, *g*, and *h* as follows.

$$\begin{aligned}\int_a^b \mathbf{r}(t)\,dt &= \lim_{n\to\infty}\sum_{i=1}^{n}\mathbf{r}\left(t_i^*\right)\Delta t \\ &= \lim_{n\to\infty}\left[\left(\sum_{i=1}^{n} f\left(t_i^*\right)\Delta t\right)\mathbf{i}+\left(\sum_{i=1}^{n} g\left(t_i^*\right)\Delta t\right)\mathbf{j}+\left(\sum_{i=1}^{n} h\left(t_i^*\right)\Delta t\right)\mathbf{k}\right]\end{aligned}$$

Integrals

So

$$\int_a^b \mathbf{r}(t)\,dt = \left(\int_a^b f(t)\,dt\right)\mathbf{i} + \left(\int_a^b g(t)\,dt\right)\mathbf{j} + \left(\int_a^b h(t)\,dt\right)\mathbf{k}$$

This means that we can evaluate an integral of a vector function by integrating each component function.

Integrals

We can extend the Fundamental Theorem of Calculus to continuous vector functions as follows:

$$\int_a^b \mathbf{r}(t)\,dt = \mathbf{R}(t)\Big]_a^b = \mathbf{R}(b) - \mathbf{R}(a)$$

where **R** is an antiderivative of **r**, that is, $\mathbf{R}'(t) = \mathbf{r}(t)$.

We use the notation $\int \mathbf{r}(t)\,dt$ for indefinite integrals (antiderivatives).

Example 4

If $\mathbf{r}(t) = 2\cos t\,\mathbf{i} + \sin t\,\mathbf{j} + 2t\,\mathbf{k}$, then

$$\begin{aligned}\int \mathbf{r}(t)\,dt &= \left(\int 2\cos t\,dt\right)\mathbf{i} + \left(\int \sin t\,dt\right)\mathbf{j} + \left(\int 2t\,dt\right)\mathbf{k} \\ &= 2\sin t\,\mathbf{i} - \cos t\,\mathbf{j} + t^2\mathbf{k} + \mathbf{C}\end{aligned}$$

where $\mathbf{C}$ is a vector constant of integration, and

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \mathbf{r}(t)\,dt &= \left[2\sin t\,\mathbf{i} - \cos t\,\mathbf{j} + t^2\,\mathbf{k}\right]_0^{\frac{\pi}{2}} \\ &= 2\mathbf{i} + \mathbf{j} + \frac{\pi^2}{4}\mathbf{k}\end{aligned}$$